

```
import pandas as pd

data = {
    'Product': ['A', 'B', 'C', 'D', 'E'],
    'Sales': [1200, 1500, 900, 1800, 1300],
    'Profit': [300, 450, 200, 600, 350],
    'Advertising': [100, 120, 80, 150, 110]
}

df = pd.DataFrame(data)
print(df)
```

	Product	Sales	Profit	Advertising
0	A	1200	300	100
1	B	1500	450	120
2	C	900	200	80
3	D	1800	600	150
4	E	1300	350	110

```
import matplotlib.pyplot as plt

avg_sales = df['Sales'].mean()
print("Average Sales:", avg_sales)

print("\nSummary Statistics:")
print(df.describe())

plt.figure(figsize=(6,4))
plt.bar(df['Product'], df['Sales'])
plt.title('Sales by Product')
plt.xlabel('Product')
plt.ylabel('Sales')
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df['Advertising'], df['Sales'])
plt.title('Advertising vs Sales')
plt.xlabel('Advertising')
plt.ylabel('Sales')
plt.show()

corr_matrix = df[['Sales', 'Profit', 'Advertising']].corr()

plt.figure(figsize=(6,4))
plt.imshow(corr_matrix)
plt.colorbar()
plt.xticks(range(len(corr_matrix.columns)), corr_matrix.columns)
plt.yticks(range(len(corr_matrix.columns)), corr_matrix.columns)
plt.title('Correlation Heatmap')
plt.show()

print("\nCorrelation Matrix:")
print(corr_matrix)
```

Average Sales: 1340.0

Summary Statistics:

	Sales	Profit	Advertising
count	5.000000	5.000000	5.000000
mean	1340.000000	380.000000	112.000000
std	336.154726	152.479507	25.884358
min	900.000000	200.000000	80.000000
25%	1200.000000	300.000000	100.000000
50%	1300.000000	350.000000	110.000000
75%	1500.000000	450.000000	120.000000
max	1800.000000	600.000000	150.000000

